

AIDOS LAB
AI FOR DATA-ORIENTED SCIENCE

UNI FR

Lecture 7: (More) persistent homology

Geometry and Topology in Machine Learning

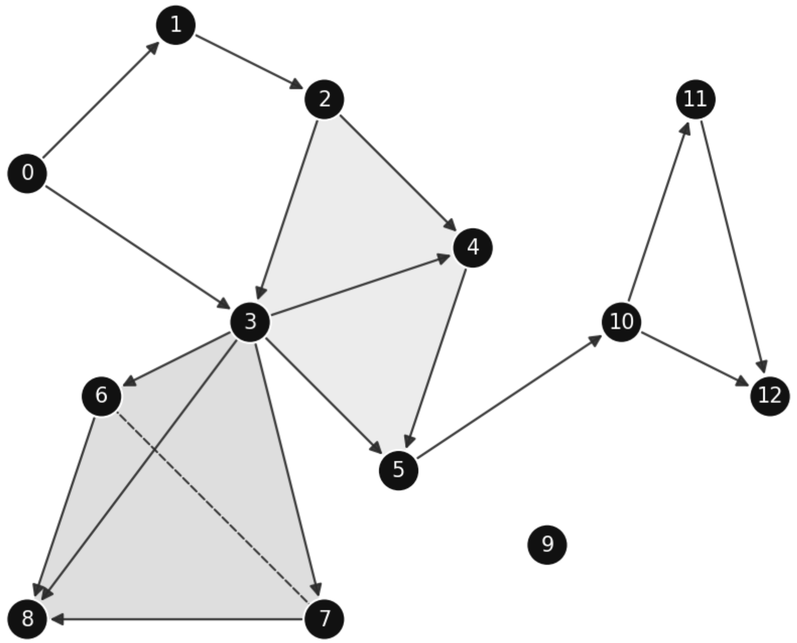
Inés García Redondo — October 27, 2026

Recap

Lecture 7 — (More) persistent homology

We now know how to count holes

Chains and boundaries



- **Chain space:** let $k \geq 0$, the k-chain space $C_k(X)$ is the *vector space* spanned by the *oriented simplices*

$$C_k(X) = \text{span}\{(i_0, \ldots, i_k) : i_0 < \cdots < i_k\} \cong \mathbb{R}^{|X_k|}$$

$$c = \sum_{\sigma \in X_k} a_\sigma \cdot \sigma, \quad a_\sigma \in \mathbb{R}$$

- **Boundary operator:** *linear map* connecting chain spaces of consecutive dimension $\partial_k : C_k(X) \to C_{k-1}(X)$ defined on the basis elements as

$$\partial_k(i_0, \ldots, i_k) = \sum_{j=0}^{k} (-1)^j \ (i_0, \ldots, \widehat{i_j}, \ldots i_k)$$

Example:

$\partial_3(3,6,7,8) =$

We now know how to count holes

Betti numbers

Def. Homology group

Cycles

$$H_k(X) = \frac{\ker \partial_k}{\operatorname{im} \partial_{k+1}}$$

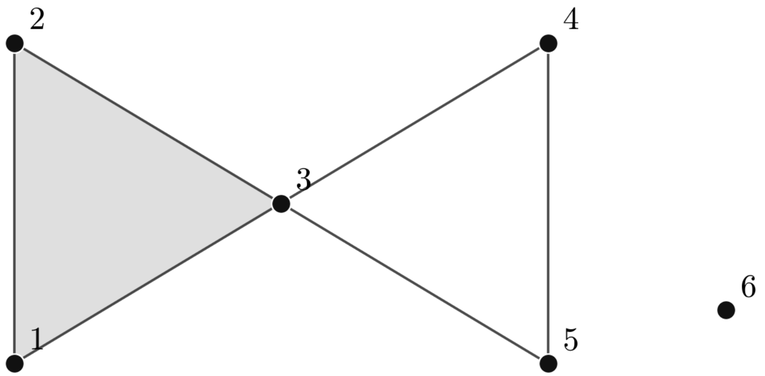
Boundaries

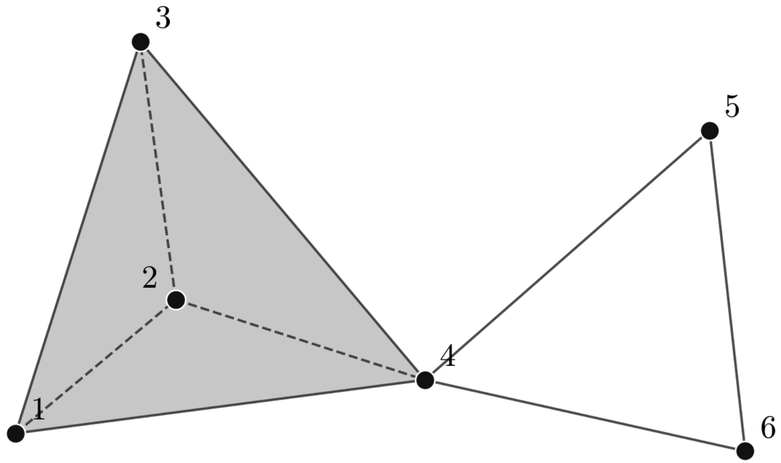
Def. Betti number

$$\beta_k(X) = \dim H_k(X)$$

"Number of k-dimensional holes in X"

Try it: What are the Betti numbers?





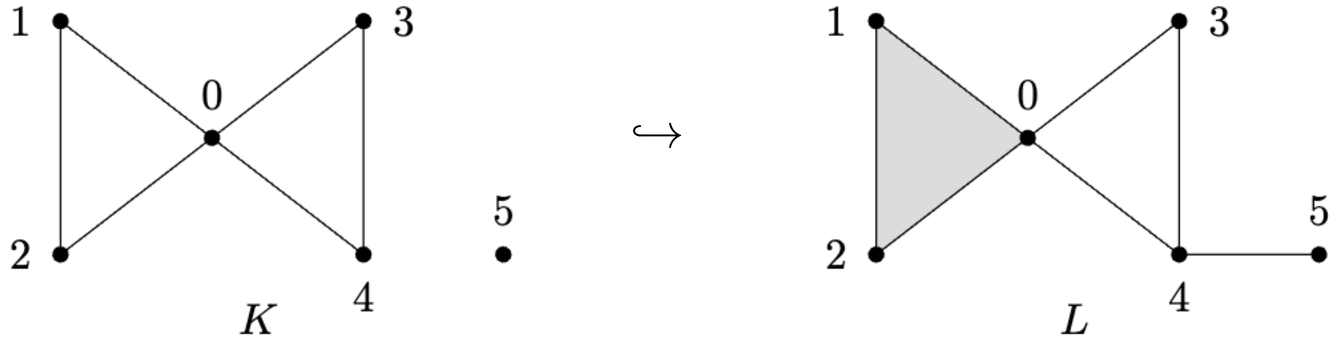
Induced maps

Def. Induced map at the chain level

A simplicial map $f : X \to Y$ induces a linear map $f_\# : C_k(X) \to C_k(Y)$ for all $k \geq 0$:

$$\alpha = \sum_{\sigma \in X_k} a_\sigma \cdot \sigma \mapsto f_\#(\alpha) = \sum_{\sigma \in X_k : f(\sigma) \in Y_k} a_\sigma \cdot f(\sigma)$$

Try it: What are the matrices of the induced maps I_0, I_1, I_2?



Induced maps

Def. Induced map at the homology level

A simplicial map $f : X \to Y$ induces a linear map $f_* : H_k(X) \to H_k(Y)$ for all $k \geq 0$:

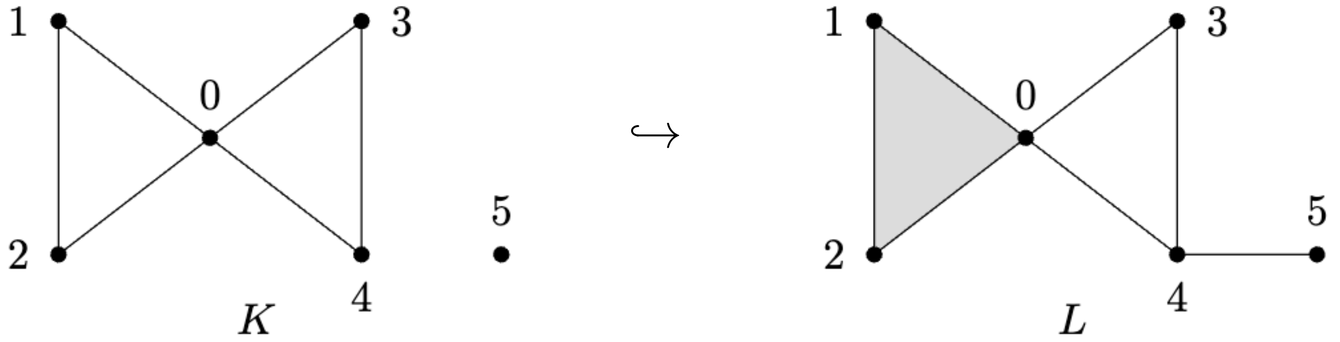
$$[\alpha] = \{\alpha + \partial_{k+1}\beta : \beta \in C_{k+1}(X)\} \mapsto f_*([\alpha]) = \big[f_\#(\alpha)\big] = \big\{f_\#(\alpha) + \partial_k^Y \beta \ : \ \beta \in C_{k+1}(Y)\big\}$$

Try it: What are the matrices of the induced maps in homology with bases

	K	L
H_0	$\{[0]_K, [5]_K\}$	$\{[0]_L\}$
H_1	$\{[a]_K, [b]_K\}$	$\{[b]_L\}$.

$a = (0,1) - (0,2) + (1,2)$

$b = (0,3) - (0,4) + (3,4)$



Induced maps

Two important properties

Properties of the induced maps

The induced homology map $f_* : H_k(X) \to H_k(Y)$ for all $k \geq 0$ satisfies the following two properties:

- If $\mathrm{id} : X \to X$ is the identity map, sending each vertex in X to itself, then $\mathrm{id}_* : H_k(X) \to H_k(X)$ is the identity map of vector spaces, which leaves every vector unchanged
- If $f : X \to Y$ and $g : Y \to Z$ then

$$(g \circ f)_* = g_* \circ f_*$$

Induced maps

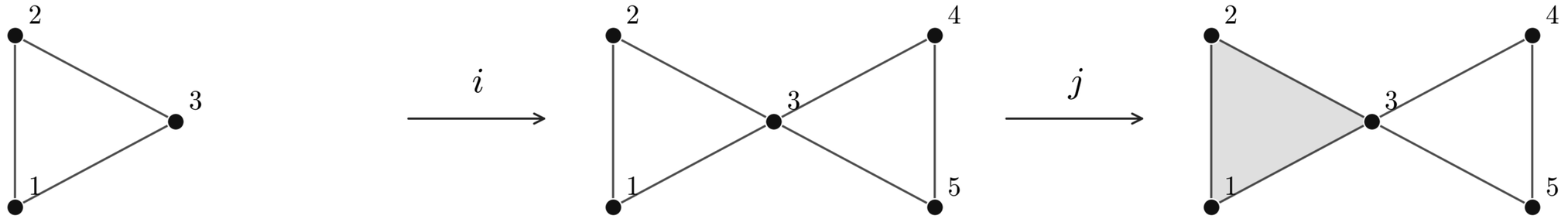
Two important properties

Properties of the induced maps

Induces maps in homology satisfy:

- $\mathrm{id}_* = \mathrm{id}_{H_k(X)}$
- $(g \circ f)_* = g_* \circ f_*$

K L M



$H_1(K) = \langle [a] \rangle$ $H_1(L) = \langle [a], [b] \rangle$ $H_1(M) = \langle [b] \rangle$

Motivation

What we want to solve today

Suppose you are given the following point cloud, how can you design a test to conclude that this sample comes from a distribution over *two circles*?





The nerve theorem

Lecture 7 — (More) persistent homology

Recap: Čech complex

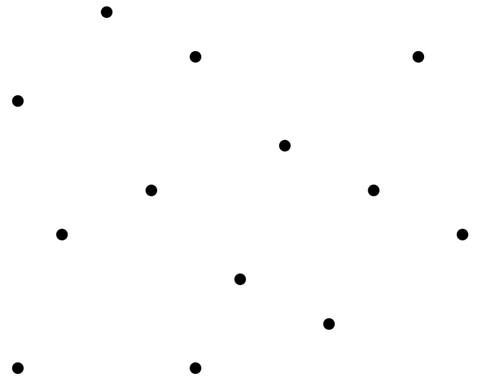
Finite metric space

Given (S, d) and $r \geq 0$, the **Čech complex** (or Nerve complex) at scale r is:

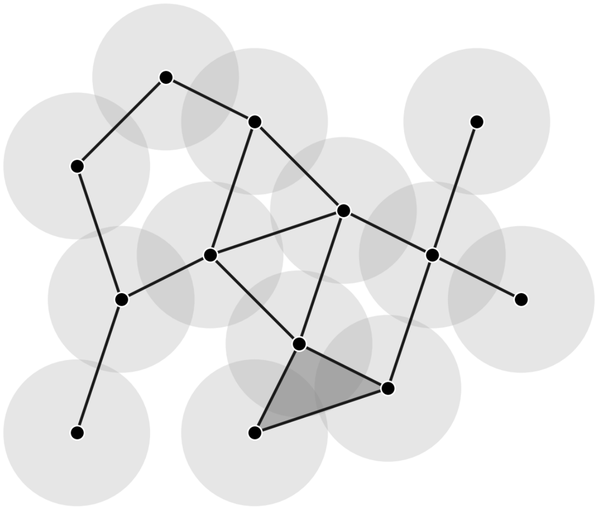
$$C_r(S, d) = \left\{ \{s_{i_0}, \ldots, s_{i_k}\} : \bigcap_{j=0}^{k} B(s_{i_j}, r) \neq \emptyset \right\}$$

Monotonicity

For $r \leq s$: $C_r(S) \subseteq C_s(S)$



Point cloud



Čech complex

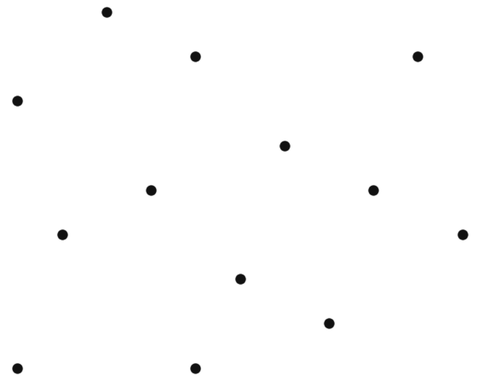
Recap: Vietoris—Rips complex

Given (S, d) and $r \geq 0$, the **Vietoris-Rips complex** at scale r is:

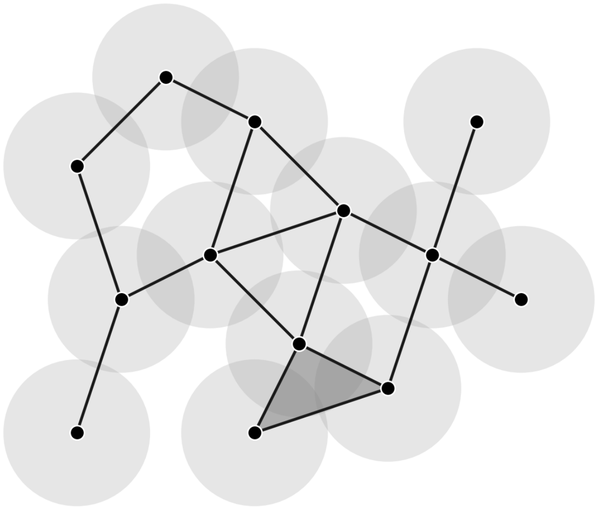
$$VR_r(S, d) = \left\{ \{s_{i_0}, \ldots, s_{i_k}\} : d(s_i, s_j) \leq 2r,\ \forall i_0 \leq i < j \leq i_k \right\}$$

Monotonicity

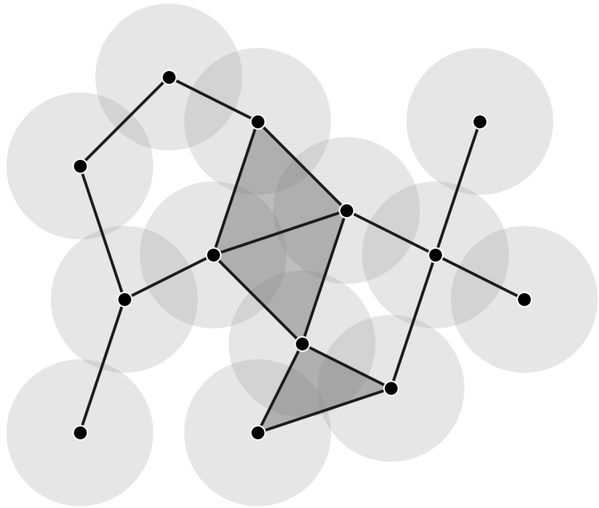
For $r \leq s$: $VR_r(S) \subseteq VR_s(S)$



Point cloud



Čech complex



VR complex

Nerve theorem

Def + thm. Nerve theorem (informal)

Let X be a topological space and $\mathcal{U} = \{B(x, r) : x \in S \subset X,\ r > 0\}$ a *cover* of this space, that is

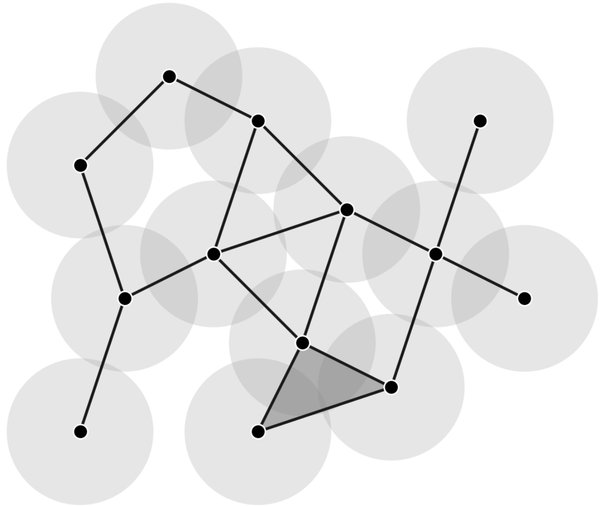
$$X \subset \bigcup_{x \in S} B(x, r)$$

The *nerve* of this cover is the following simplicial complex:

$$\text{Nerve}(\mathcal{U}) = \left\{ \sigma \subset S : \bigcap_{x \in \sigma} B(x, r) \neq \varnothing \right\}$$

With some *niceness assumptions* on X and $\mathcal{U}$ we have:

$$\text{Nerve}(\mathcal{U}) \cong X \qquad \textbf{(homotopy equivalent, same Betti numbers)}$$



The *Čech complex* is a nerve

$$C_r(S, d) = \left\{ \{s_{i_0}, \ldots, s_{i_k}\} : \bigcap_{j=0}^{k} B(s_{i_j}, r) \neq \varnothing \right\}$$

[1]Ghrist, R., and Muhammad, A. "Coverage and Hole-Detection in Sensor Networks via Homology." *IPSN 2005. Fourth International Symposium on Information Processing in Sensor Networks*, 2005., April 2005, 254–60.

[1]Bauer, U., Kerber, M., Roll, F., and Rolle, A. "A Unified View on the Functorial Nerve Theorem and Its Variations." *Expositiones Mathematicae* 41, no. 4 (2023): 125503.

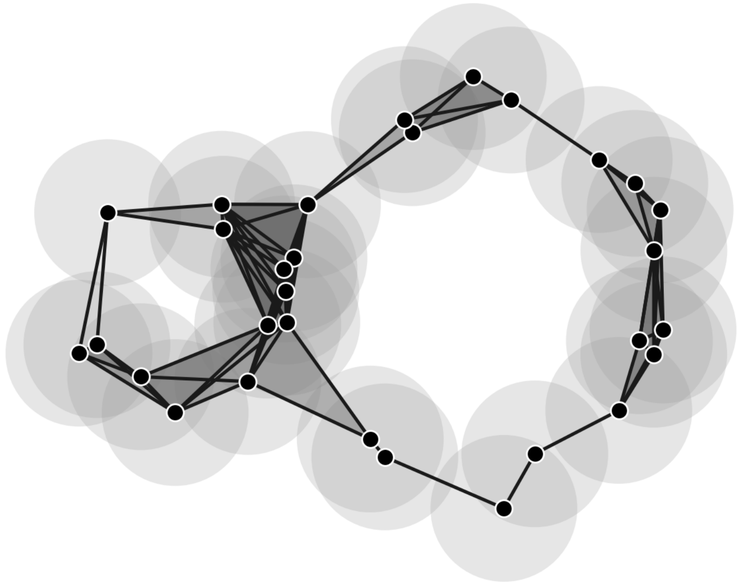
Nerve theorem: practical consequences

Suppose you are given the following point cloud, how can you design a test to conclude that this sample comes from a distribution over *two circles*?



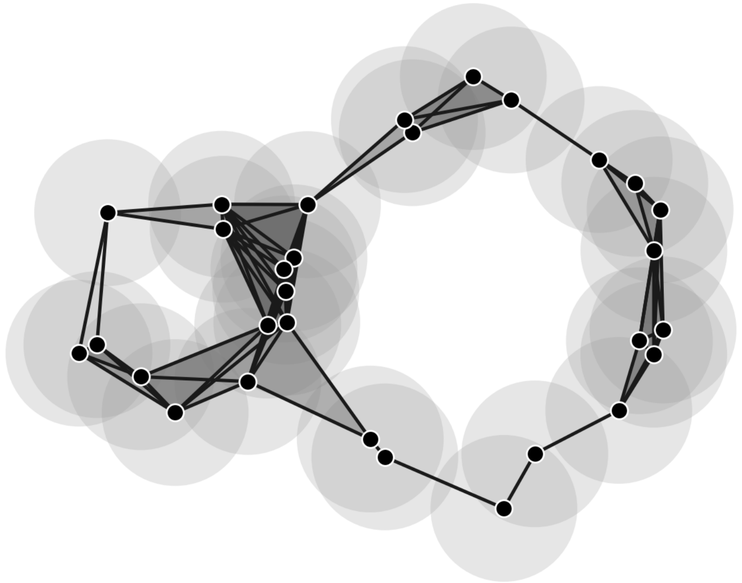
Nerve theorem: practical consequences

1. Approximate S with a Čech complex for a some radius that captures the holes: $C_r(S)$
2. Compute homology in degree 1: $H_1(C_r(S))$
3. If $\beta_1(C_r(S)) = 2$, then you are observing two holes



Nerve theorem: practical consequences

1. Approximate S with a Čech complex for a some radius that captures the holes: $C_r(S)$
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3. If $\beta_1(C_r(S)) = 2$, then you are observing two holes



How do you find the radius r?

Persistent homology

Lecture 7 — (More) persistent homology

Filtrations

Consider all possible radii!

<u>Def.</u> Filtration

$\{X_t : t \in \mathbb{R}\}$ such that for $s \leq t$, $X_s \subseteq X_t$

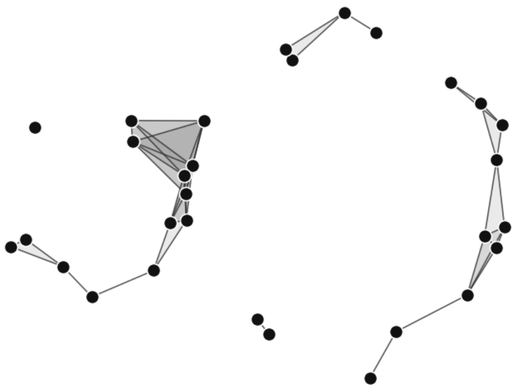
$t = 0$



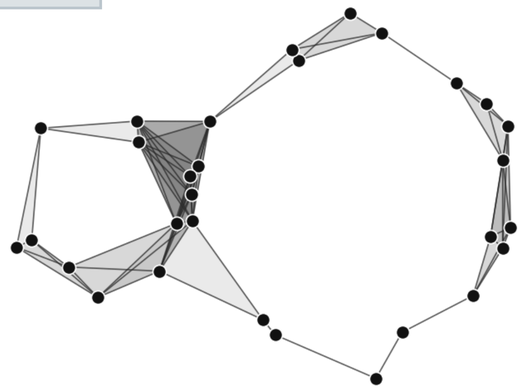
$t = 0.6$



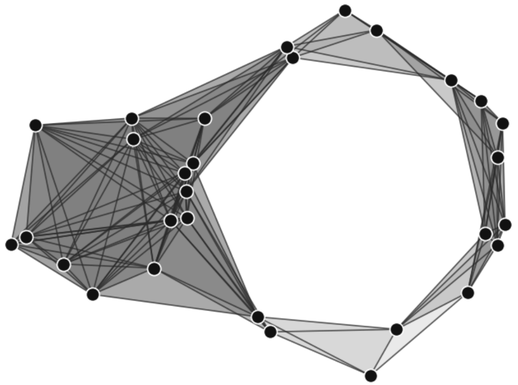
$t = 1$



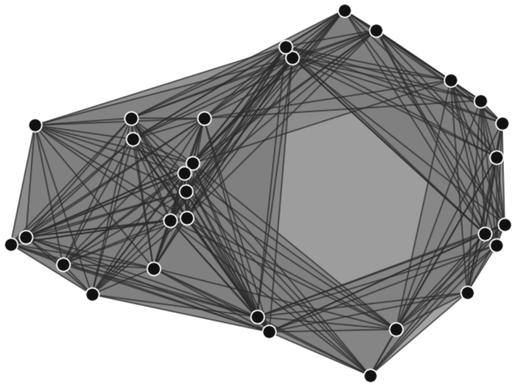
$t = 1.5$



$t = 2.2$



$t = 3.4$



Persistent homology module

From the property about compositions and induced maps!

Persistent homology module:

$$X_\bullet : X_1 \hookrightarrow X_2 \hookrightarrow \ldots \hookrightarrow X_N \quad \rightsquigarrow$$

Linear maps: induced maps from the inclusion maps

$$H_k(X_\bullet) : H_k(X_1) \xrightarrow{f_2^1} H_k(X_2) \xrightarrow{f_3^2} \ldots \xrightarrow{f_N^{N-1}} H_k(X_N)$$

Vector spaces $\cong \mathbb{R}^{|(X_i)_k|}$

with $X_l \subseteq X_n \subseteq X_m$

$$f_m^n \circ f_n^l = f_m^l$$

Persistence module

Def. Persistence module (in full generality)

A family $\{V_n\}_{1\leq n\leq N}$ of vector spaces and a family $\{\phi_m^n : V_n \to V_m\}_{n\leq m}$ of linear maps satisfying:

- $\phi_n^n = \mathrm{id}_{V_n}$
- $\phi_n^m \circ \phi_m^l = \phi_n^l$ for all $l \leq m \leq n$

An example of persistence modules are interval modules:

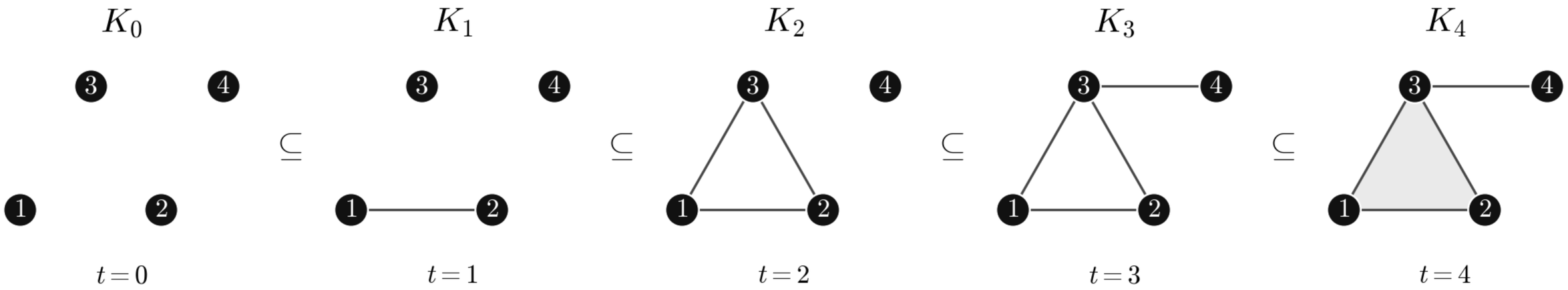
$$I[a,b]_n = \begin{cases} 0 & \text{if } n < a, \text{ or } n > b \\ \mathbb{R} & \text{if } a \leq n \leq b \end{cases} \qquad I[a,b]_m^n = \begin{cases} 0 & \text{if } n < a, \text{ or } m > b \\ \mathrm{id}_{\mathbb{R}} & \text{if } a \leq n \leq m \leq b \end{cases}$$

Module: $0 \to 0 \to \cdots \to 0 \longrightarrow \mathbb{R} \xrightarrow{\mathrm{id}_{\mathbb{R}}} \mathbb{R} \longrightarrow \cdots \longrightarrow \mathbb{R} \xrightarrow{\mathrm{id}_{\mathbb{R}}} \mathbb{R} \longrightarrow 0 \to \cdots \to 0$

Parameters: $1 \quad 2 \qquad a-1 \quad a \quad a+1 \qquad b-1 \quad b \quad b+1 \qquad N$

Persistent homology

An example

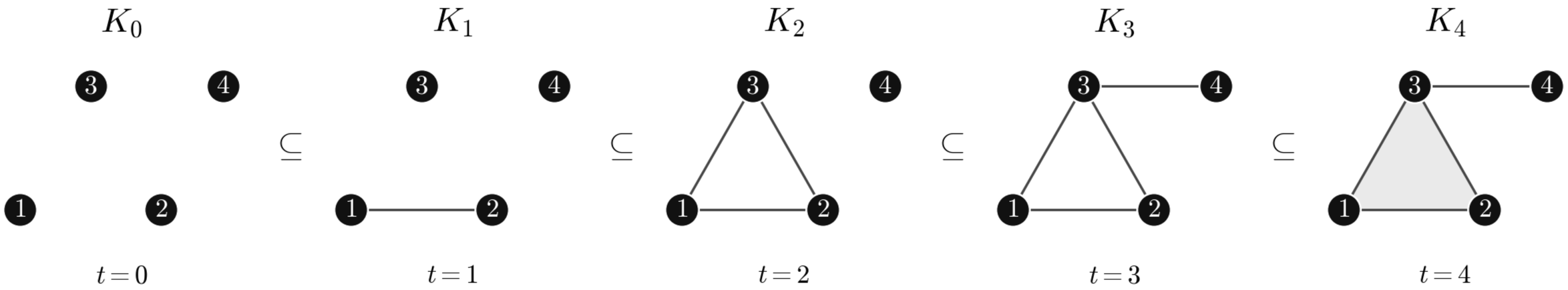


For 0 and 1-dimensional homology:

- Choose generators for the homology groups at each step in the filtration
- Connect the generators in subsequent steps in the filtration whenever the map induced in homology by the inclusion connects them
- No line needs to be drawn when the generator is directly mapped to 0 by the induced map

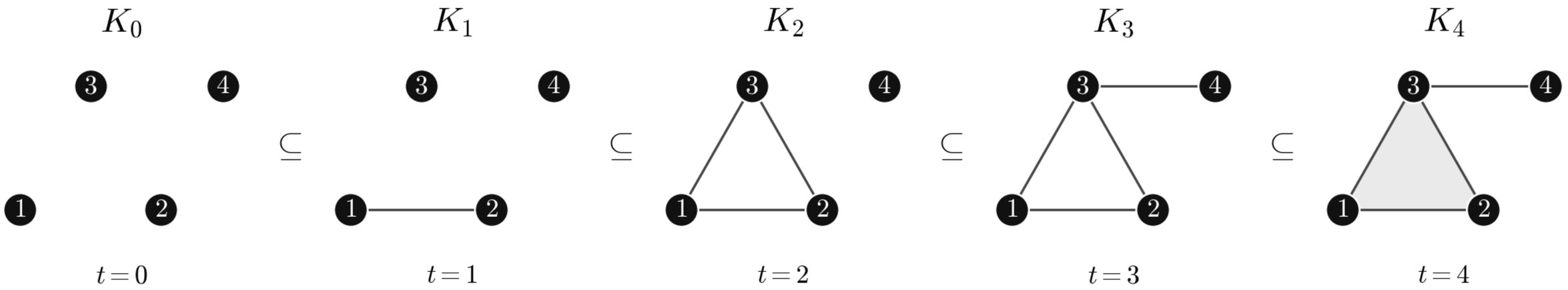
Persistent homology

$H_0(K_\bullet)$



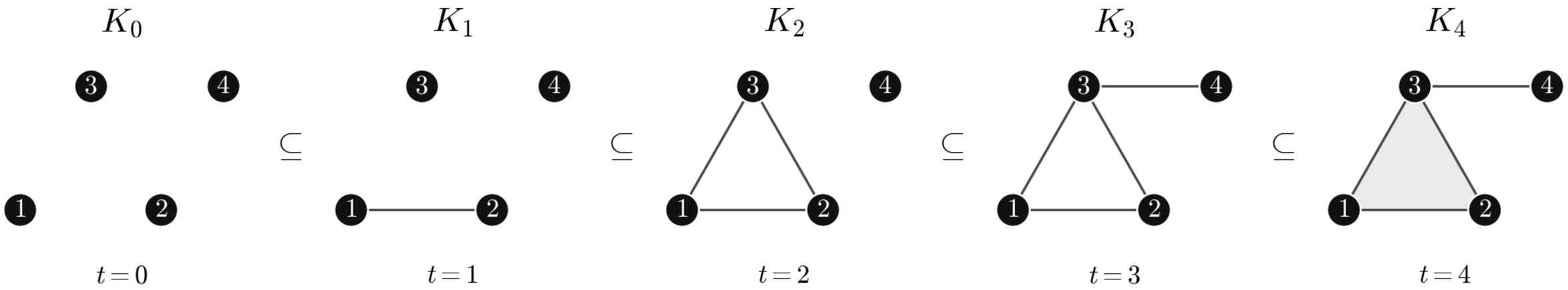
Persistent homology

$H_1(K_\bullet)$



Persistent homology

Choosing generators more wisely

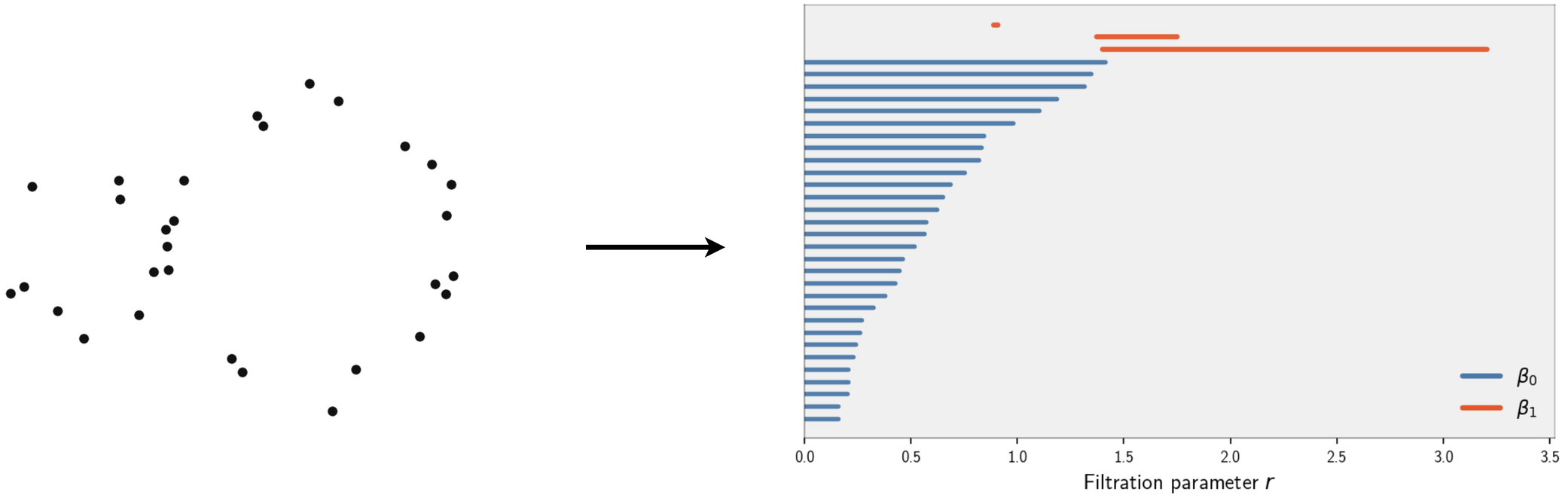


Persistence barcode

This is true in general by the **structure theorem**, whose statement and proof go beyond the scope of the course

It is enough that you know that for each filtration (and corresponding persistent homology module) we can compute a **barcode** that tracks how homology classes are created and destroyed as the filtration grows

This is the same as tracking the evolution of the Betti numbers of the complexes in the filtration

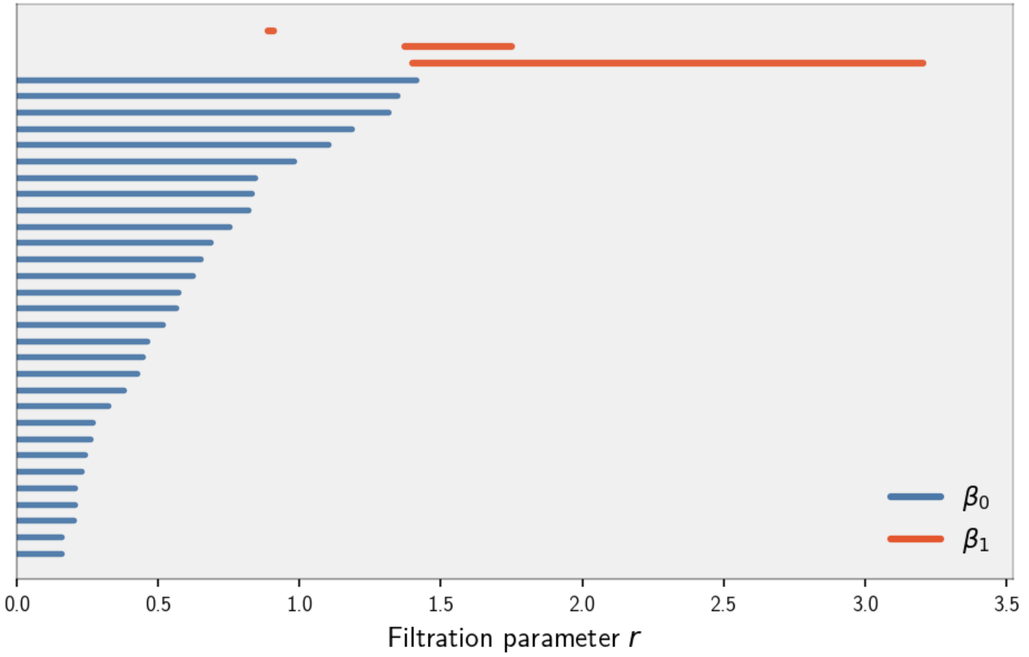


Azumaya, Gorô. "Corrections and Supplementaries to My Paper Concerning Krull-Remak-Schmidt's Theorem." *Nagoya Mathematical Journal* 1 (June 1950): 117–24.

Crawley-Boevey, William. "Decomposition of Pointwise Finite-Dimensional Persistence Modules." *Journal of Algebra and Its Applications* 14, no. 05 (2015): 1550066.

Persistence barcode

As many bars as points, bars *die* as points get connected

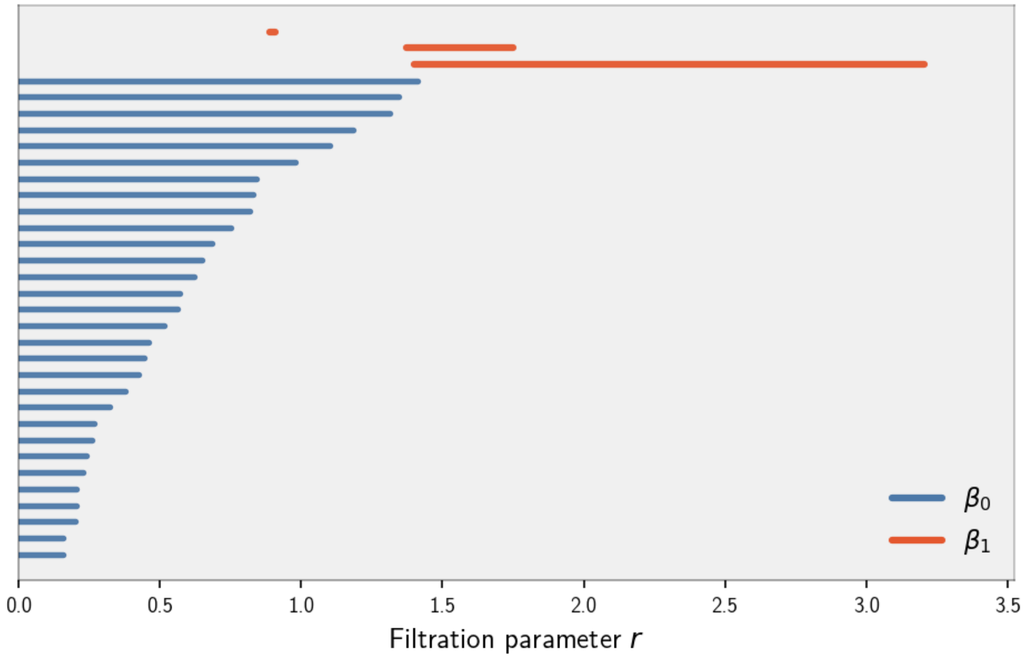


Two long bars: we have two *persistent* loops in the point cloud

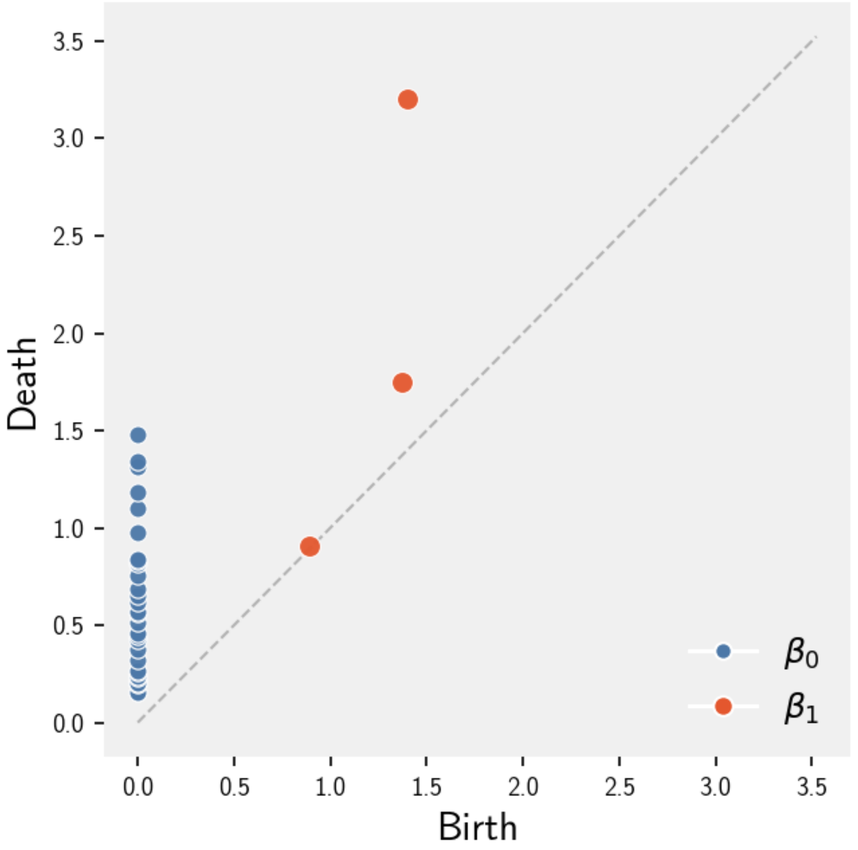
The point cloud had the shape of two circles!

Persistence barcode

Persistence diagram



Persistence barcode



Persistence diagram

Computing barcodes

Lecture 7 — (More) persistent homology

Filtration boundary matrix

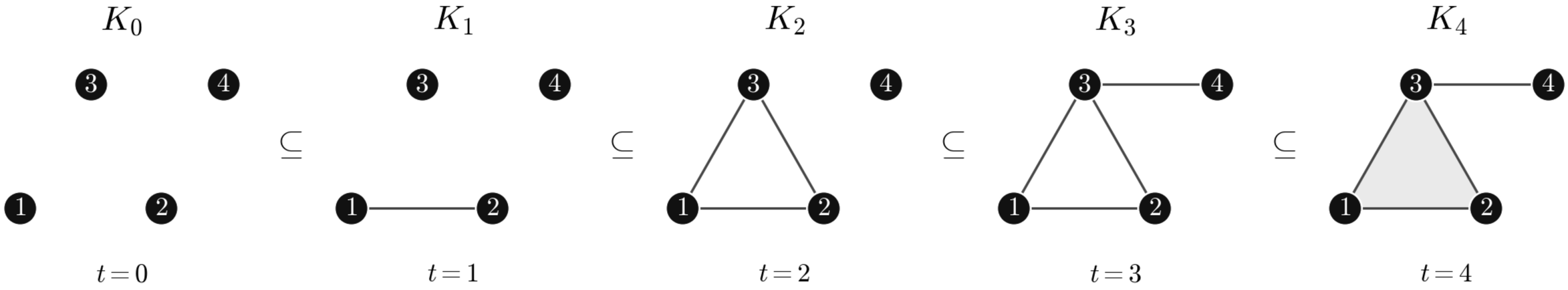
How to define it?

- Order all simplices in the simplicial complex at the last step of the filtration $K = K_N$ by increasing filtration value
- When several simplices enter the filtration at the same time, order them by dimension
- Break ties arbitrarily when two simplices of the same dimension enter the filtration at the same time
- Compute the boundary matrices D_k representing the boundary maps $\partial_k : C_k(K) \to C_{k-1}(K)$ of the simplicial complex at the last step of the filtration, where all simplices have been added
- Write the full boundary matrix as the block matrix given by all these matrices

$$D = \bigoplus_{k=0}^{\dim K} D_k$$

- D_{ij} = coefficient of σ_i in $\partial\sigma_j$ with indices ranging all the simplices in K

Reducing the boundary matrix



Reducing the boundary matrix

Reduction algorithm

Input: boundary matrix of a filtration D

Steps:

$R \leftarrow D$

$n \leftarrow \#$ columns of R

For $1 \leq j \leq n$:

  If $R_j \neq 0$:

    pivot$(R_j) \leftarrow \max\{i : R_{ij} \neq 0\}$

    For $1 \leq k < j$:

      If pivot$(R_k) =$ pivot(R_j):

        $R_j \leftarrow R_j \pm R_k$ so that pivot(R_j) increases

Output: reduced matrix R

Example

$$D = \left(\begin{array}{c|cccc|cccc|c} & 1 & 2 & 3 & 4 & 12 & 13 & 23 & 34 & 123 \\ \hline 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 3 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ \hline 12 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 13 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 23 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 34 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 123 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

Reading off the barcode

- If pivot$(R_j) = i$: σ_i creates a homology class and σ_j kills it. We have to add a *finite bar*

$$[f(\sigma_i), f(\sigma_j))$$

  where $f(\sigma_i)$ = filtration value at which σ_i enters the filtration, in dimension $\dim(\sigma_i)$

- If $R_j = 0$ (and there is no pivot), that means that this column represents a cycle that never gets killed, so we get an *essential bar*

$$[f(\sigma_j), +\infty)$$

  in dimension $\dim(\sigma_j)$

Example

The idea behind the algorithm

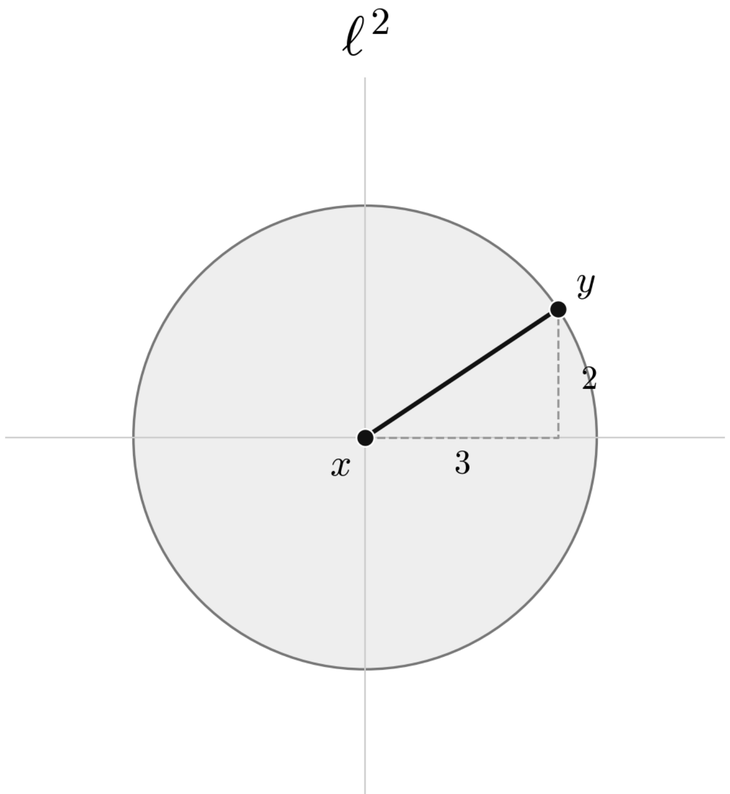
Notice that we are doing exactly the same thing as we did before:

- Starting from the basis of simplices in each dimension (basis of the chain space)
- We add of subtract simplices that have already appeared in the filtration to reduce the boundary, that is, *we change the element of the basis over which we are computing the boundary*, adding the vectors of the basis that we already had
- We do this until we obtain a *boundary that is linearly independent* with respect to the other boundaries already present, or until we arrive to the zero column (the combination of simplices is a *cycle*)
- This process gives us a basis from which we can read the barcode, just as before!

Distances in diagrams

Lecture 7 — (More) persistent homology

Euclidean distance in $\mathbb{R}^2$



$$d_2(x,y) = \sqrt{3^2 + 2^2} = \sqrt{13}$$

$$x = (0,0), \quad y = (3,2)$$

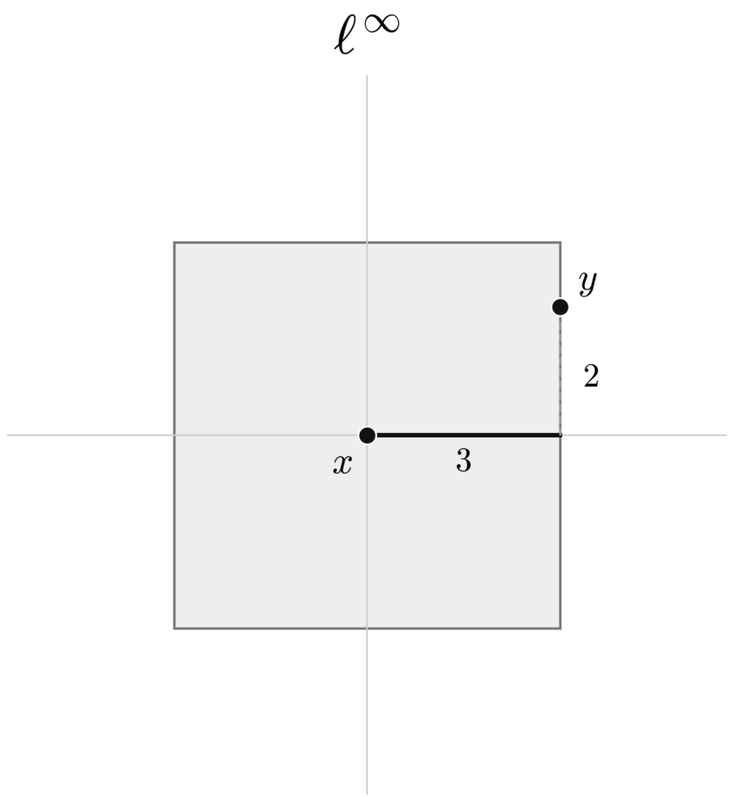
We know how to compute distances in $\mathbb{R}^2$

$$d(x,y) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2}$$

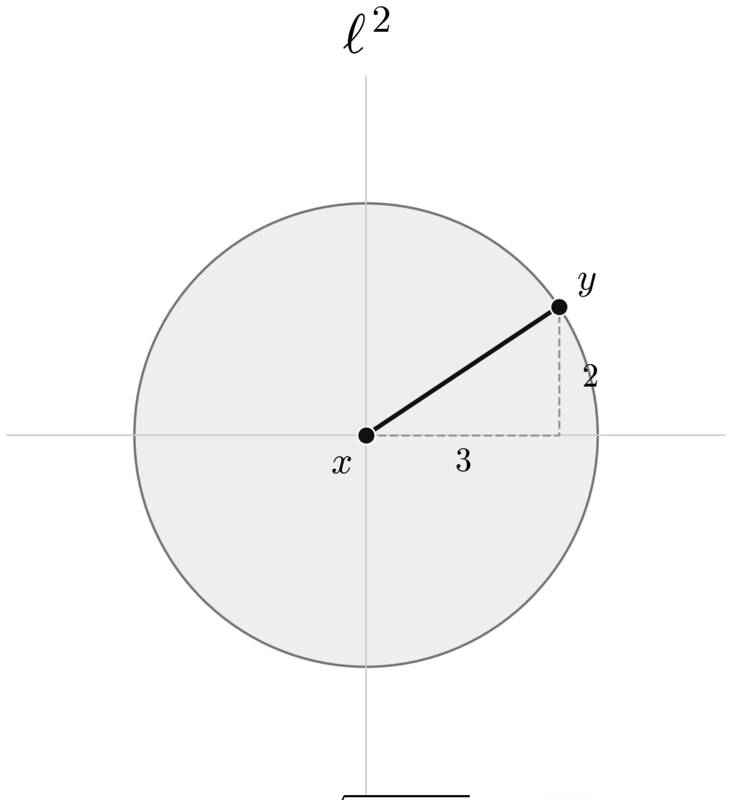
$$B(x,r) = \{y \in \mathbb{R}^2 : d(x,y) \leq r\}$$

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2}$$

ℓ^p distances in $\mathbb{R}^2$

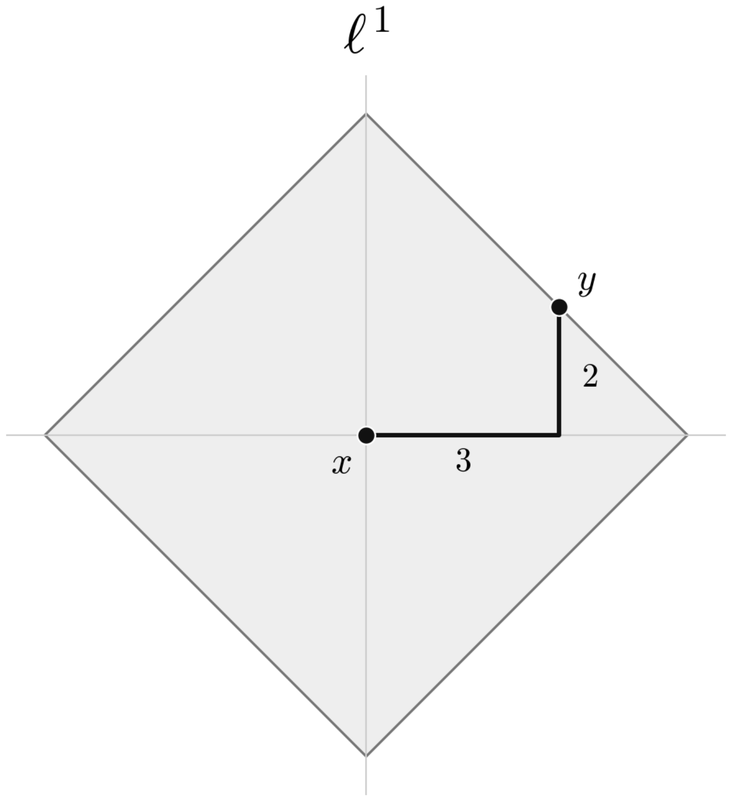


$$d_\infty(x,y) = \max\{3,2\} = 3$$



$$d_2(x,y) = \sqrt{3^2+2^2} = \sqrt{13}$$

$$x = (0,0), \quad y = (3,2)$$



$$d_1(x,y) = 3+2 = 5$$

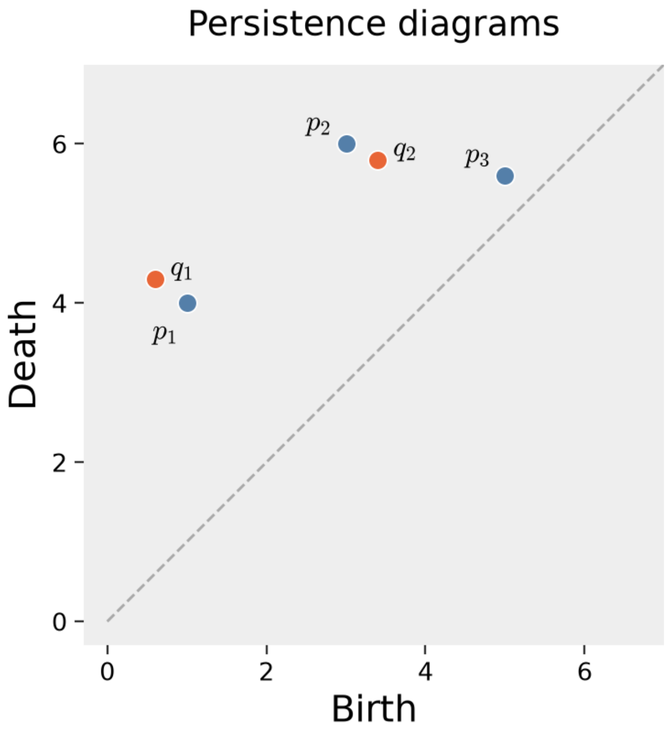
$$d_p(x,y) = \left(|x_1 - y_1|^p + |x_2 - y_2|^p \right)^{1/p}$$

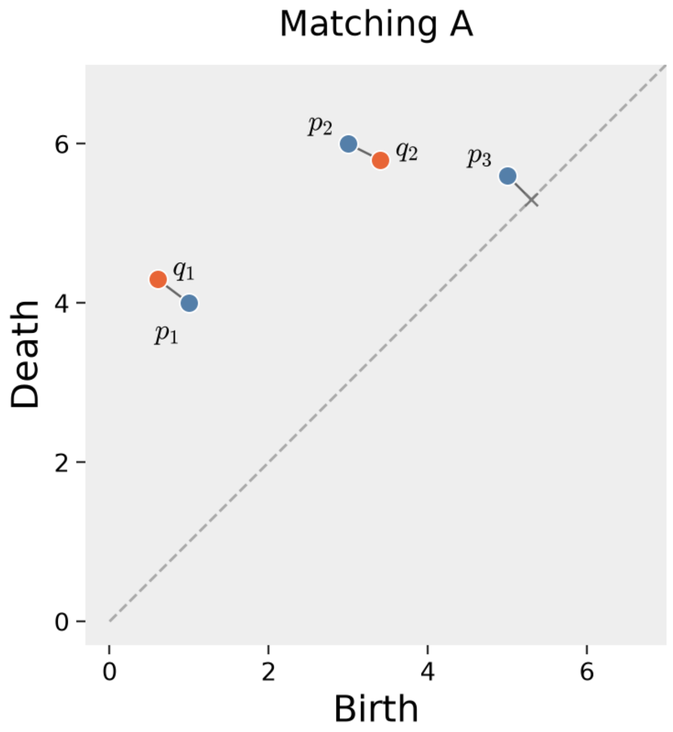
$$B_p(x,r) = \{y \in \mathbb{R}^2 : d_p(x,y) \le r\}$$

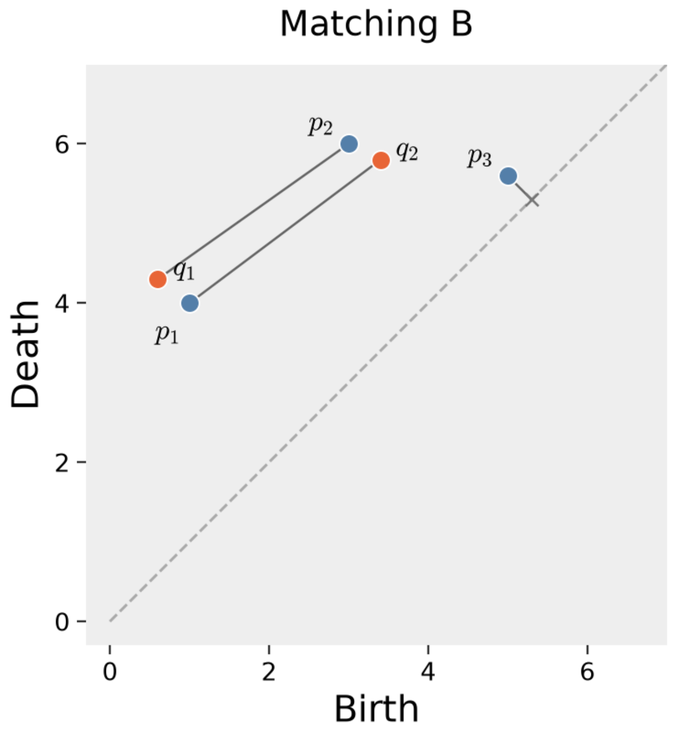
Matchings between diagrams

Def. Matching between persistence diagrams

A matching between persistence diagrams is a bijection $\psi : \mathrm{PD}(X) \to \mathrm{PD}(Y)$ between the points in the diagrams, where the points in the diagonal in both diagrams are counted with infinite multiplicity







Distances between diagrams

Def. Matching between persistence diagrams

A matching between persistence diagrams is a bijection $\psi : \mathrm{PD}(X) \to \mathrm{PD}(Y)$ between the points in the diagrams, where the points in the diagonal in both diagrams are counted with infinite multiplicity

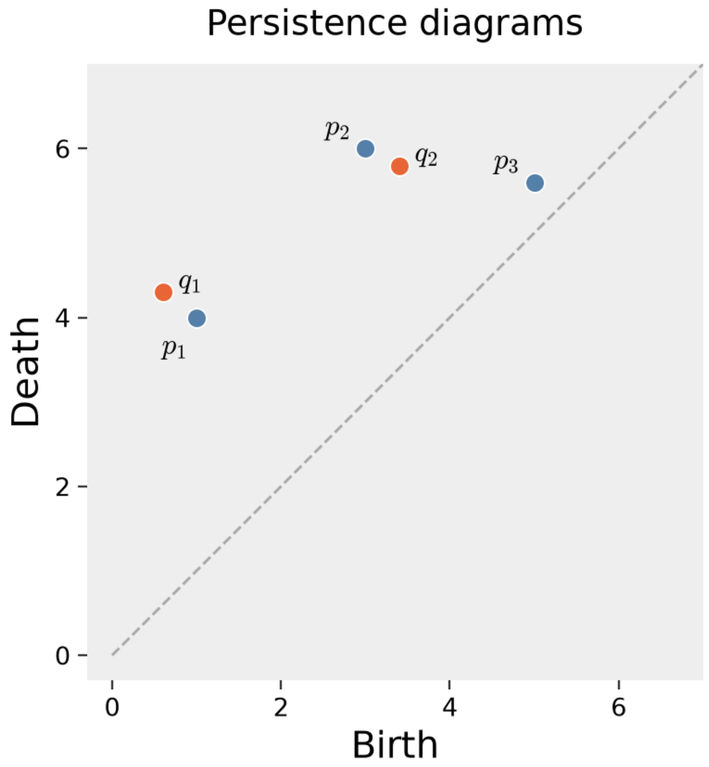
Bottleneck distance

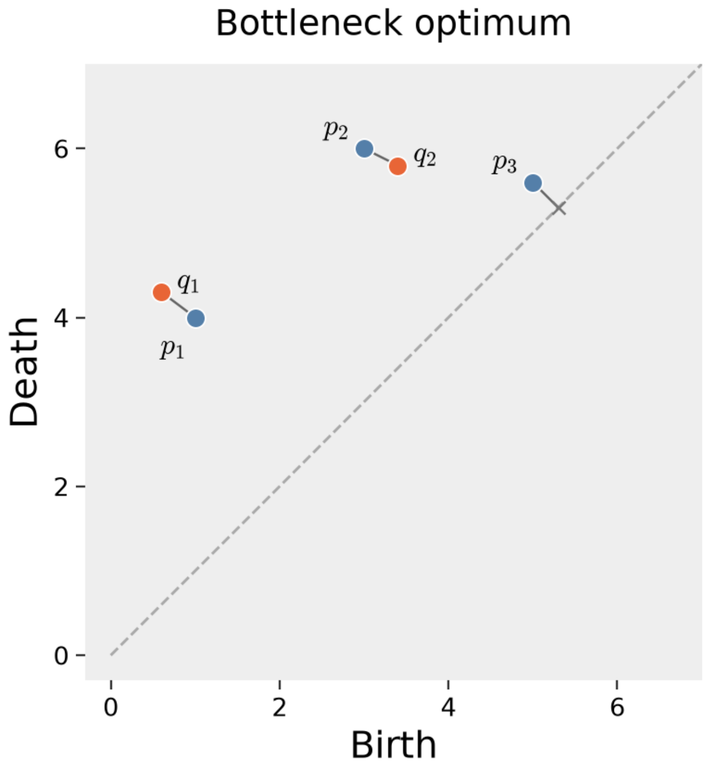
$$d_B\left(\mathrm{PD}(X), \mathrm{PD}(Y)\right) = \inf_{\psi:\mathrm{PD}(X)\to\mathrm{PD}(Y)} \max\{\|x - \psi(x)\|_\infty : \forall x \in \mathrm{PD}(X)\}$$

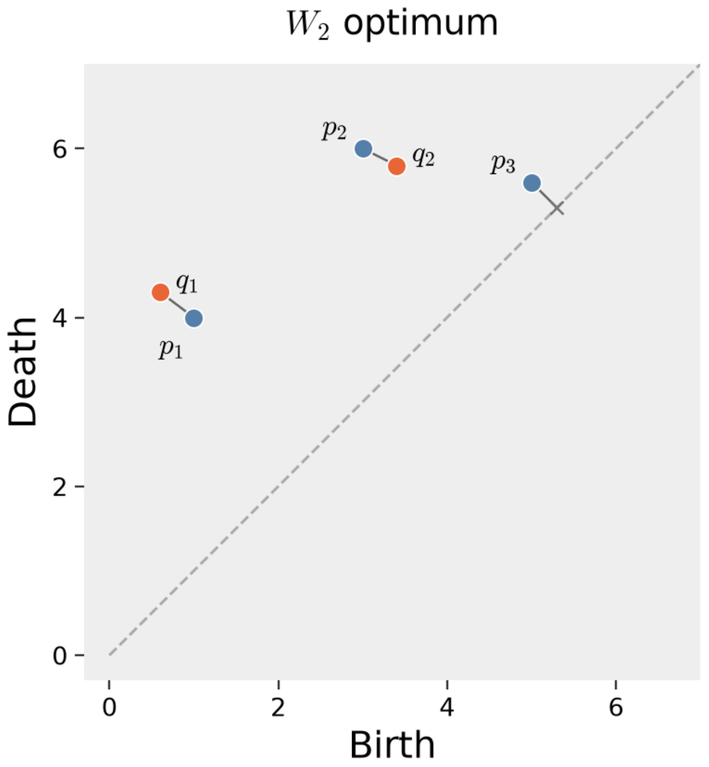
Wasserstein distance

$$W_p\left(\mathrm{PD}(X), \mathrm{PD}(Y)\right) = \inf_{\psi:\mathrm{PD}(X)\to\mathrm{PD}(Y)} \left(\sum_{x\in\mathrm{PD}(X)} \|x - \psi(x)\|_p^p \right)^{1/p}$$

Distances between diagrams





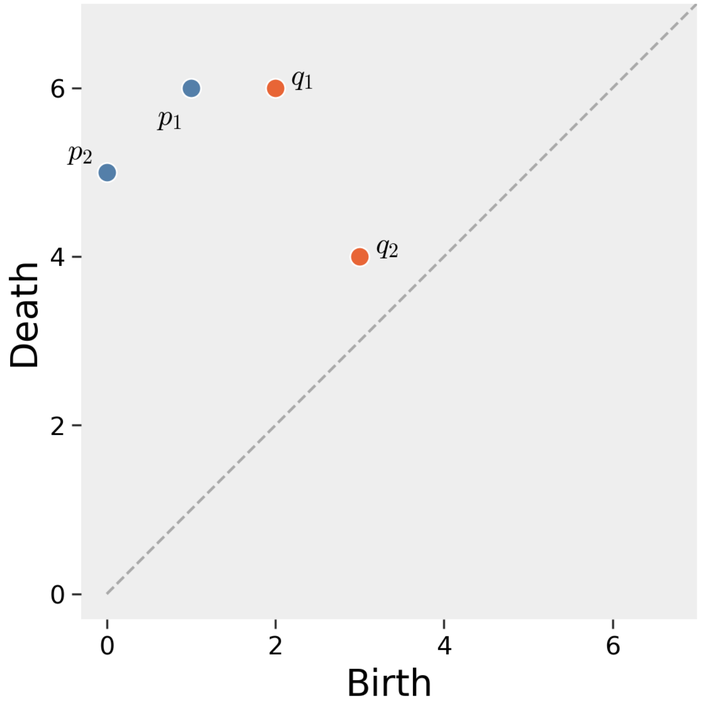


$$d_B = \max \|x - \psi(x)\|_\infty = 0.4$$

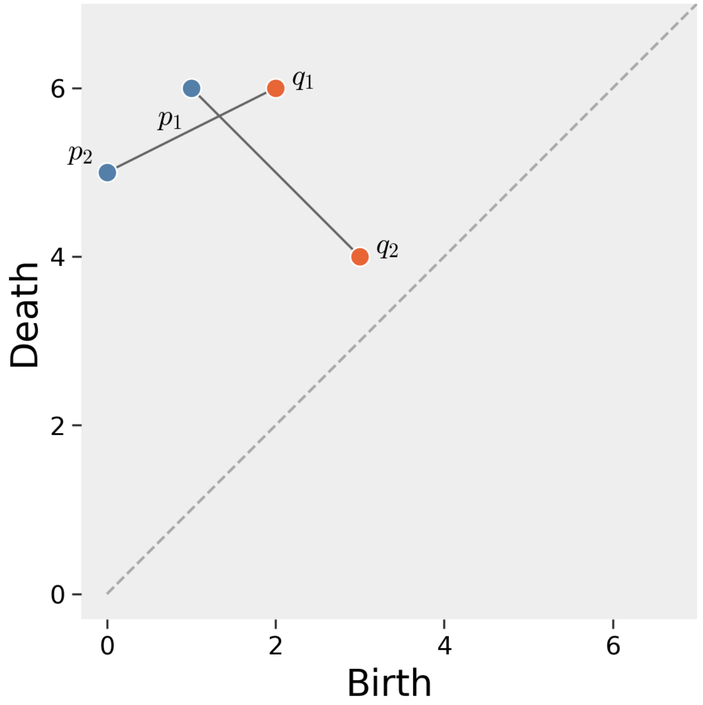
$$W_2 = \left(\sum \|x - \psi(x)\|_2^2\right)^{1/2} = 0.7937$$

Distances between diagrams

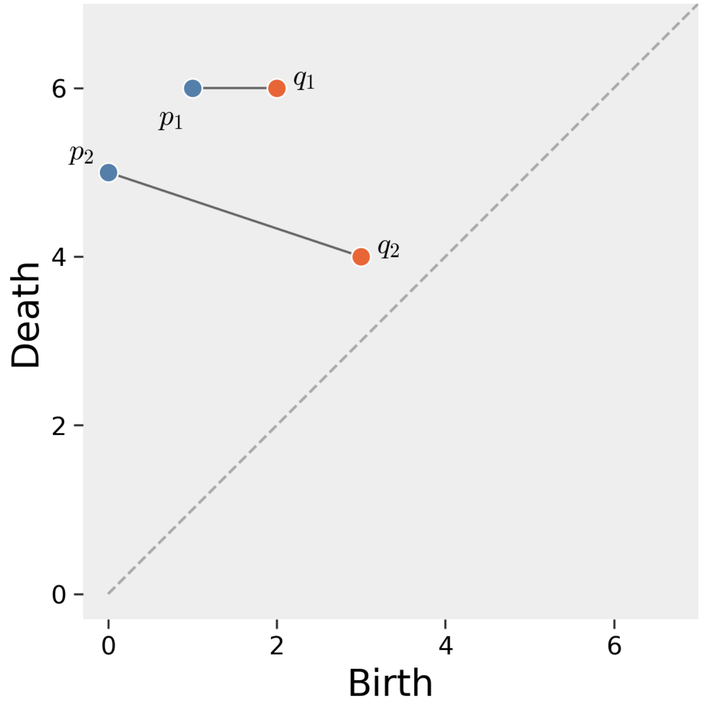
Persistence diagrams



Bottleneck optimum



W_2 optimum



$$d_B = \max \|x - \psi(x)\|_\infty = 2$$

$$W_2 = \left(\sum \|x - \psi(x)\|_2^2\right)^{1/2} = 3.3166$$

To sum up

Today we have learnt how to...

- How to define the persistence module
- How to represent a persistence module as a persistence barcode or a persistence diagram
- How to compute barcodes
- How to compute distances between persistence diagrams

Thank you for your attention!



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Lecture 7 — (More) Persistent Homology

Inés García Redondo — October 27, 2026